

23CSE111 OBJECT ORIENTED PROGRAMMING

S2 B.Tech CSE

Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – I (Analysis)

Problem Identification & Class Mapping (CO1)

Group No : D6

Team Members :

Roll No	Student Name
AM.SC.U4CSE25346	S.V Karthik
AM.SC.U4CSE25343	Vedananda Reddy Ramireddy
AM.SC.U4CSE25309	Bendalam Sai Ashish
AM.SC.U4CSE25333	Mullangi Puri Jagannadh

1. Problem Title : Online Shopping System

2. Problem Description: Retail businesses require digital storefronts to manage products, carts, and orders efficiently. This system provides an interactive academic e-commerce simulation where customers can register, log in, and browse products using a filtering service. The system automates billing by calculating totals dynamically and generating a bill PDF upon order placement, while storing order history for future reference

3. Identified Entities (Classes)

Class Name	Description
Customer	Manages user profile, authentication credentials, and personal order history.
Product	Represents the catalog items with metadata like price and category for
Cart	A stateful container that tracks selected items and calculates real-time totals.
Order	Encapsulates a finalized transaction, including the date and generated bill PDF.
UserAuthService	Manages security aspects like user registration and login.
ProductFilterService	Handles logic for searching and filtering products by categories.
OrderService	Orchestrates checkout, generates PDFs, and saves order history.

4. Class Attributes and Behaviours

Class Name	Attributes	Methods (behaviours)
Customer	customerID, username, email,	register(), login(), viewOrderHistory()
Product	productID, name, category,	getDetails(), isAvailable(), updateStock()
Cart	cartID, itemsList, subTotal	addItem(), removeItem(),
Order	orderID, customerID,	generateBillPDF(), printSummary()
ProductFilterService	allProducts, activeFilters	fetchItems(), applyCategoryFilter(), searchByName()
OrderService	transactionID, billingLogic	processCheckout(), saveToHistory()